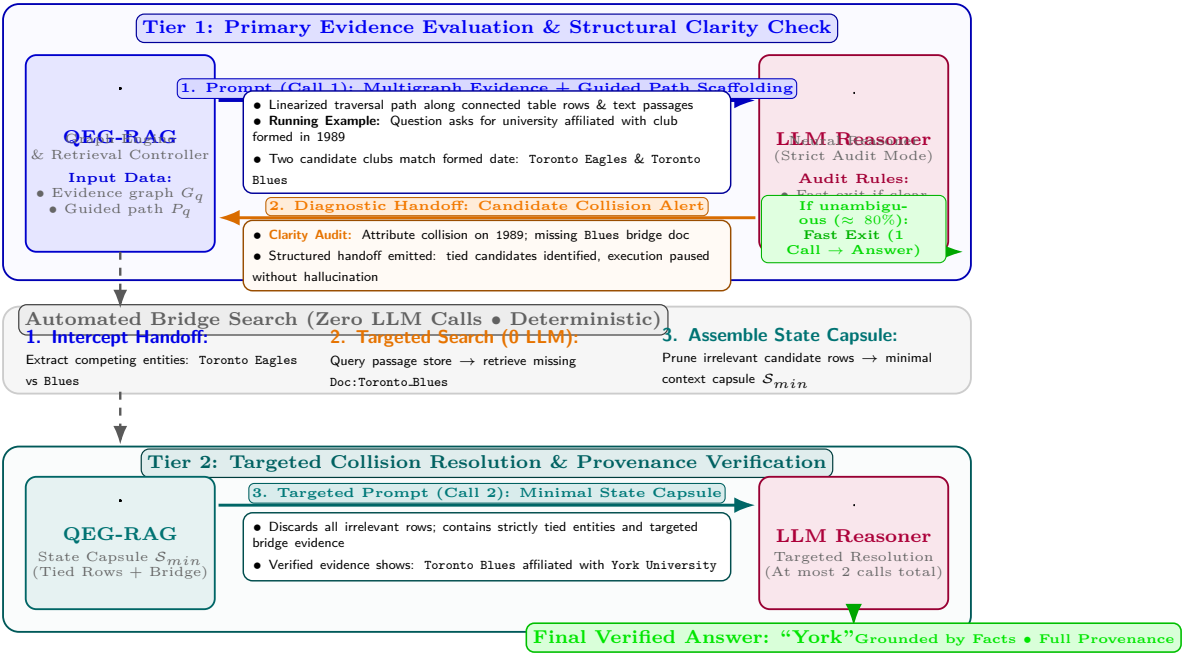

Tier 1: Primary Evidence Evaluation & Structural Clarity Check
QEG-RAG
& Retrieval Controller
Input Data:
• Evidence graph $G_q$
• Guided path $P_q$
1. Prompt (Call 1): Multigraph Evidence + Guided Path Scaffolding
• Linearized traversal path along connected table rows & text passages
• Running Example: Question asks for university affiliated with club formed in 1989
• Two candidate clubs match formed date: Toronto Eagles & Toronto Blues
LLM Reasoner
(Strict Audit Mode)
Audit Rules:
2. Diagnostic Handoff: Candidate Collision Alert
• Clarity Audit: Attribute collision on 1989; missing Blues bridge doc
• Structured handoff emitted: tied candidates identified, execution paused without hallucination
If unambiguous (≈ 80%): Fast Exit (1 Call → Answer)
Automated Bridge Search (Zero LLM Calls • Deterministic)
1. Intercept Handoff:
Extract competing entities: Toronto Eagles vs Blues
2. Targeted Search (0 LLM):
Query passage store → retrieve missing Doc:Toronto_Blues
3. Assemble State Capsule:
Prune irrelevant candidate rows → minimal context capsule $\mathcal{S}_{min}$
Tier 2: Targeted Collision Resolution & Provenance Verification
QEG-RAG
State Capsule $\mathcal{S}_{min}$
(Tied Rows + Bridge)
3. Targeted Prompt (Call 2): Minimal State Capsule
• Discards all irrelevant rows; contains strictly tied entities and targeted bridge evidence
• Verified evidence shows: Toronto Blues affiliated with York University
LLM Reasoner
Targeted Resolution
(At most 2 calls total)
Final Verified Answer: "York" Grounded by Facts • Full Provenance